The genus-1 seven faces: elliptic collision residues and the KZB connection

Raeez Lorgat

Abstract

At genus 0, the collision residue of a chirally Koszul chiral algebra admits seven equivalent presentations: as a twisting morphism, a line-operator r-matrix, a λ-bracket, commuting sphere Hamiltonians, the Drinfeld r-matrix, the Sklyanin bracket, and the Gaudin Hamiltonians. We prove the genus-1 extension: on an elliptic curve E_τ, the genus-1 collision residue $r^{(1)}_{\mathcal{A}}(z,\tau)$ is the elliptic regularization of the genus-0 residue, obtained by replacing rational propagator kernels with Weierstrass functions. All seven faces at genus 1 specialize a *single E_1-datum* $\Theta_{\mathcal{A}}$ via the dictionary $\Psi \rightsquigarrow \hbar \rightsquigarrow \tau$: Ψ is the algebraic spectral parameter on genus-0 faces, $\hbar$ is the quantisation parameter on the Drinfeld / Sklyanin / Yangian faces, and τ is the elliptic modulus on the KZB / Belavin–Drinfeld / Felder faces. The passage is the bar-intrinsic projection $\Theta^{(g=1)}_{\mathcal{A}} = \mathrm{Reg}_\tau \circ \Theta^{(g=0)}_{\mathcal{A}}$ where Reg_τ is elliptic regularisation. For affine Kac–Moody algebras $\mathcal{A} = \widehat{\mathfrak{g}}_k$, the seven faces specialize to the Knizhnik–Zamolodchikov–Bernard connection, the Belavin–Drinfeld elliptic r-matrix, the Felder dynamical elliptic $R(z,\tau)$ (via Fay trisecant), and the elliptic Gaudin Hamiltonians. The B-cycle monodromy of the collision residue recovers the quantum group parameter $q = \exp(2\pi i/(k+h^\vee))$. For class M algebras (Virasoro, $\mathcal{W}_N$), the collision residue produces higher-order elliptic differential operators involving $\wp, \wp', \ldots, \wp^{(2N-3)}$ that have no counterpart in the existing integrable systems literature.

Remark 0.1 (Wave-14 reconstitution anchor). All seven faces at genus 1 factor through a single E_1-datum, namely the universal Maurer–Cartan element $\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathfrak{g}^{E_1}_{\mathcal{A}})$ (Vol I thm:recursive-existence). The face-specific specialisations are:

- *KZB / Bernard face.* $(dz, d\tau)$ components via the elliptic regularisation $1/z^{n+1} \mapsto (-1)^{n-1}\wp^{(n-1)}(z,\tau)/n!$ at $n \geq 1$, $1/z \mapsto \zeta_\tau(z)$.
- *Felder dynamical face.* The elliptic dynamical $R(z,\tau)$ replaces the pentagon of genus-0 coherence by the Fay trisecant identity (Vol I theorem status: chiral QG equiv elliptic, thm:chiral-qg-equiv-elliptic).
- *Belavin–Drinfeld face.* $r^{\mathrm{ell}}_{\mathfrak{g}}(z,\tau)/(k+h^\vee)$ with theta-function ratios $\phi_\pm(z,\tau)$ (4.3).
- *Gaudin / Hitchin face.* Elliptic Hitchin on $\mathrm{Bun}_G(E_\tau)$ restricted to trivial bundle; the commutators close via the elliptic infinitesimal braid relation.

The dictionary $\Psi \rightsquigarrow \hbar \rightsquigarrow \tau$ is the passage from the algebraic genus-0 presentation to the analytic genus-1 presentation; all three parameters are bookkeeping projections of the same $\Theta_{\mathcal{A}}$. Convention note: we specify Weierstrass ζ_τ vs θ_1'/θ_1 explicitly; we use ζ_τ throughout with Legendre identity $\eta_1\tau - \eta_\tau = \pi i$. The period prefactor is $q = e^{2\pi i\hbar}$, not $e^{\pi i\hbar}$; substitute integer k to verify $|q| = 1$.

1 Introduction

The genus-0 seven-face theorem [8] identifies the collision residue $r_{\mathcal{A}}(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}})$ of a chirally Koszul chiral algebra $\mathcal{A}$ with seven classical objects: a bar-cobar twisting morphism, a line-operator r-matrix in the Dimofte–Niu–Py framework [4], a classical λ-bracket in the Khan–Zeng 3d sigma-model [6], the Gaiotto–Zeng commuting sphere Hamiltonians, the Drinfeld r-matrix, the Sklyanin Poisson bracket, and the Gaudin Hamiltonians. The seven presentations agree as elements of $\mathrm{End}_{\mathcal{A}}(2)[\![z^{-1}]\!]$; their mutual identification is controlled by the collision depth $k_{\max}$ and the shadow class $G/L/C/M$.

The collision residue $r_{\mathcal{A}}(z)$ is the degree-2 component of the E_1 Maurer–Cartan element $\Theta^{E_1}_{\mathcal{A}}$ in the ordered convolution algebra $\mathfrak{g}^{E_1}_{\mathcal{A}}$, where $B^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ is the cofree tensor coalgebra with deconcatenation coproduct. Its Σ_n-coinvariant projection is the scalar shadow κ: $\mathrm{av}(r(z))$ in abelian and scalar families, and $\mathrm{av}(r(z)) + \dim(\mathfrak{g})/2$ for non-abelian affine Kac–Moody; the spectral parameter that κ discards is precisely the data the seven faces read. At genus 1, the spectral parameter acquires modular dependence: $r^{(1)}_{\mathcal{A}}(z,\tau)$ depends on the elliptic modulus τ, and the B-cycle monodromy of the ordered collision residue encodes the quantum group parameter.

This paper extends the identification to genus 1. The elliptic curve $E_\tau = \mathbb{C}/(\mathbb{Z} + \mathbb{Z}\tau)$ replaces the Riemann sphere, the rational propagator $1/z$ is replaced by the Weierstrass zeta function $\zeta_\tau(z)$, and each rational function $1/z^{n+1}$ in the collision expansion is replaced by its canonical elliptic counterpart. The resulting genus-1 collision residue $r^{(1)}_{\mathcal{A}}(z,\tau)$ admits seven equivalent presentations that degenerate to the genus-0 faces as $\tau \to i\infty$.

The main results are:

(1) The *elliptic regularization theorem* (Theorem 2.4): the genus-1 collision residue is

$$r^{(1)}_{\mathcal{A}}(z,\tau) \;=\; c_0\,\zeta_\tau(z) \;+\; \sum_{n\geq 1} \frac{(-1)^{n-1}}{n!}\, c_n \cdot \wp^{(n-1)}(z,\tau),$$

where $c_n = a_{(n)}b$ are the OPE-mode collision coefficients.

(2) The identification of the dz-component of the *KZB connection* with the genus-1 collision residue, and of the $d\tau$-component with its z-derivative (Theorem 3.1).

(3) The identification of the genus-1 collision residue with the *Belavin–Drinfeld elliptic r-matrix* (Theorem 4.1).

(4) The *elliptic Gaudin commutativity*: the elliptic Gaudin Hamiltonians commute as a consequence of the classical elliptic Yang–Baxter equation (Theorem 5.1).

(5) The *class M collision residue* for Virasoro and $\mathcal{W}_N$: higher-order elliptic operators that are genuinely new (Theorems 6.1 and 6.3).

(6) The *degeneration theorem*: all seven genus-1 faces degenerate continuously to their genus-0 counterparts as $q = e^{2\pi i\tau} \to 0$ (Theorem 7.1).

The paper is organized as follows. Section 2 introduces the genus-1 bar propagator and the elliptic collision residue. Section 3 proves the KZB identification (Face 4). Section 4 proves the elliptic r-matrix identification (Face 5). Section 5 proves the elliptic Gaudin commutativity (Face 7). Section 6 computes the class M collision residue for Virasoro and $\mathcal{W}_N$. Section 7 establishes the degeneration to genus 0. Section 8 describes the computational verification.

2 The elliptic collision residue

2.1 The genus-1 bar propagator

At genus 0 the bar propagator is $\eta_{12} = d\log(z_1 - z_2)$: a meromorphic one-form on $\mathbb{P}^1$ with a simple pole along the diagonal. On the elliptic curve E_τ, Liouville's theorem forbids a meromorphic function with a single simple pole that is doubly periodic. The propagator must be replaced by a quasi-periodic object.

Definition 2.1 (Genus-1 bar propagator)**.** The *genus-1 bar propagator* on E_τ is the logarithmic derivative of the prime form:

$$\eta_{12}^{E_\tau} \;=\; d\log E(z_1, z_2) \;=\; \zeta_\tau(z_1 - z_2)\, d(z_1 - z_2), \tag{2.1}$$

where $E(z_1, z_2) = \theta_1(z_1 - z_2 \mid \tau)/\theta_1'(0 \mid \tau)$ is the prime form on E_τ (a section of $K^{-1/2} \boxtimes K^{-1/2}$), θ_1 is the Jacobi theta function, and ζ_τ is the Weierstrass zeta function for periods $(1, \tau)$:

$$\zeta_\tau(z) \;=\; \frac{1}{z} + \sum_{(m,n)\neq(0,0)} \left[\frac{1}{z - m - n\tau} + \frac{1}{m + n\tau} + \frac{z}{(m + n\tau)^2}\right]. \tag{2.2}$$

The Weierstrass zeta function has the Laurent expansion $\zeta_\tau(z) = 1/z + O(z^3)$ near $z = 0$, matching the genus-0 propagator $1/z$ at leading order. The difference is global: ζ_τ is quasi-periodic,

$$\zeta_\tau(z + 1) = \zeta_\tau(z) + 2\eta_1, \qquad \zeta_\tau(z + \tau) = \zeta_\tau(z) + 2\eta_\tau, \tag{2.3}$$

where $\eta_1 = \zeta_\tau(1/2)$ and $\eta_\tau = \zeta_\tau(\tau/2)$ are the quasi-periods, related by Legendre's identity $\eta_1\tau - \eta_\tau = \pi i$. The quasi-periodicity of the propagator is the geometric origin of the B-cycle monodromy that distinguishes genus 1 from genus 0 in all seven faces.

Remark 2.2 (Propagator weight)*.* The bar propagator $d\log E(z, w)$ has weight 1 in both variables, regardless of the conformal weight of the fields being sewed. The logarithmic derivative of the prime form is a $(1, 0)$-form in each variable; the conformal weight h of the generator enters the OPE-mode coefficient c_n, not the propagator kernel.

2.2 The collision residue

Definition 2.3 (Genus-1 collision residue)**.** For a chirally Koszul chiral algebra $\mathcal{A}$ on E_τ, the *genus-1 collision residue* is

$$r_{\mathcal{A}}^{(1)}(z, \tau) \;:=\; \mathrm{Res}_{1,2}^{\mathrm{coll}}(\Theta_{\mathcal{A}})\big|_{E_\tau} \;\in\; \mathrm{End}_{\mathcal{A}}(2)[\![z^{-1}]\!] \otimes \mathbb{C}[\![\tau]\!], \tag{2.4}$$

where $\Theta_{\mathcal{A}}$ is the bar-intrinsic Maurer–Cartan element.

Theorem 2.4 (Elliptic regularization)**.** *Let $\mathcal{A}$ be a chirally Koszul chiral algebra with genus-0 collision residue $r_{\mathcal{A}}(z) = \sum_{n\geq 0} c_n/z^{n+1}$, where $c_n = a_{(n)}b$ are the OPE-mode coefficients. The genus-1 collision residue is*

$$r_{\mathcal{A}}^{(1)}(z, \tau) \;=\; c_0\, \zeta_\tau(z) \;+\; \sum_{n\geq 1} \frac{(-1)^{n-1}}{n!}\, c_n \cdot \wp^{(n-1)}(z, \tau), \tag{2.5}$$

where $\wp^{(m)} = d^m\wp/dz^m$. *In particular, the first three terms are*

$$r_{\mathcal{A}}^{(1)}(z,\tau) \;=\; c_0\,\zeta_\tau(z) \;+\; c_1\,\wp(z,\tau) \;-\; \tfrac{1}{2}\,c_2\,\wp'(z,\tau) \;+\; \cdots \tag{2.6}$$

The Laurent expansion of $r_{\mathcal{A}}^{(1)}$ *near* $z=0$ *agrees with* $r_{\mathcal{A}}(z)$ *at leading order.*

Proof. Each rational function $1/z^{n+1}$ $(n \geq 0)$ in the genus-0 collision expansion is the unique meromorphic function on $\mathbb{P}^1$ with a pole of order $n+1$ at $z=0$ and vanishing at $z=\infty$. On E_τ, the unique meromorphic function matching $1/z + O(z)$ at $z=0$ with the prescribed quasi-periodicity is $\zeta_\tau(z)$ (for $n=0$), while for $n \geq 1$ the unique doubly periodic function matching $1/z^{n+1} + O(z^{-n+1})$ is $(-1)^{n-1}\wp^{(n-1)}(z)/n!$. The prefactor is fixed by the Laurent expansion $\wp^{(m)}(z) = (-1)^m(m+1)!/z^{m+2} + O(1)$ at $m = n-1$:

$$\frac{(-1)^{n-1}}{n!}\,\wp^{(n-1)}(z) \;=\; \frac{(-1)^{n-1}}{n!}\cdot\frac{(-1)^{n-1}\,n!}{z^{n+1}} + O(z^{-n+1}) \;=\; \frac{1}{z^{n+1}} + O(z^{-n+1}).$$

Substituting into the propagator-extraction formula of Definition 2.1 yields (2.5). □

Remark 2.5 (Quasi-periodicity dichotomy). The $\wp$-function and all its derivatives are doubly periodic. The quasi-periodicity of $r_{\mathcal{A}}^{(1)}(z,\tau)$ is therefore inherited entirely from the ζ_τ-term:

$$r_{\mathcal{A}}^{(1)}(z+\tau,\tau) - r_{\mathcal{A}}^{(1)}(z,\tau) \;=\; 2\eta_\tau\cdot c_0. \tag{2.7}$$

When $c_0 = 0$ (the Cartan-type case: $\beta\gamma$, lattice VOAs), the collision residue is a genuine meromorphic function on E_τ. When $c_0 \neq 0$ (the Yangian-type case: affine Kac–Moody, Virasoro, $\mathcal{W}_N$), it is quasi-periodic with non-trivial B-cycle monodromy.

3 Face 4: the KZB connection

At genus 0, Face 4 identifies the collision residue with the Gaiotto–Zeng commuting sphere Hamiltonians. At genus 1, the sphere Hamiltonians are replaced by the Knizhnik–Zamolodchikov–Bernard connection on the configuration space $\mathrm{Conf}_n(E_\tau)$. The KZB connection has two components: a dz-component generalizing the genus-0 KZ connection, and a $d\tau$-component encoding the dependence of conformal blocks on the modulus τ.

Theorem 3.1 (Face 4: KZB connection)**.** *For* $\mathcal{A} = \widehat{\mathfrak{g}}_k$ *at level* $k \neq -h^\vee$, *the modular shadow connection on* $\mathrm{Conf}_n(E_\tau)$ *restricts to the KZB connection:*

$$\nabla^{\mathrm{KZB}} \;=\; d \;-\; \underbrace{\frac{1}{k+h^\vee}\sum_{i<j}\Omega_{ij}\,\zeta_\tau(z_{ij})\,dz_{ij}}_{dz\text{-component: genus-1 collision residue}} \;-\; \underbrace{\frac{1}{k+h^\vee}\sum_{i<j}\Omega_{ij}\,\wp(z_{ij},\tau)\,d\tau}_{d\tau\text{-component: modular Hamiltonian}}\;, \tag{3.1}$$

where $z_{ij} = z_i - z_j$ *and* Ω_{ij} *is the Casimir acting on the* i*-th and* j*-th tensor factors.*

(i) *The* dz*-component equals the genus-*1 *collision residue:*

$$H_{z,i}^{\mathrm{KZB}} \;=\; \frac{1}{k+h^\vee}\sum_{j\neq i}\Omega_{ij}\,\zeta_\tau(z_{ij}) \;=\; \sum_{j\neq i}\big[r_{\widehat{\mathfrak{g}}_k}^{(1)}(z_{ij},\tau)\big]_{ij}. \tag{3.2}$$

(ii) *The $d\tau$-component equals the z-derivative of the collision residue, through the Weierstrass identity $\wp(z,\tau) = -\zeta'_\tau(z)$:*

$$H^{\mathrm{KZB}}_{\tau,i} \;=\; \frac{1}{k+h^\vee}\sum_{j\neq i}\Omega_{ij}\,\wp(z_{ij},\tau) \;=\; -\sum_{j\neq i}\partial_z\big[r^{(1)}_{\widehat{\mathfrak{g}}_k}(z_{ij},\tau)\big]_{ij}. \tag{3.3}$$

Proof. **Part (i).** The affine Kac–Moody OPE $J^a(z)\,J^b(w) \sim k\,\delta^{ab}/(z-w)^2 + f^{ab}_{\ c}\,J^c(w)/(z-w)$ has $c_0 = \Omega/(k+h^\vee)$ and $c_n = 0$ for $n \geq 1$: the bar propagator absorbs one pole order, so the only surviving term is c_0. By Theorem 2.4, the elliptic regularization gives $r^{(1)}_{\widehat{\mathfrak{g}}_k}(z,\tau) = \Omega\,\zeta_\tau(z)/(k+h^\vee)$. Substituting into the Gaiotto–Zeng formula produces (3.2), which is the dz-component of the KZB connection [3, 7].

Part (ii). The $d\tau$-component involves the modular Hamiltonian D_i, whose leading term is $\sum_{j\neq i}\wp(z_{ij},\tau)\,\Omega_{ij}/(k+h^\vee)$. The Weierstrass identity $\wp(z,\tau) = -\zeta'_\tau(z)$ identifies this with $-\partial_z r^{(1)}_{\mathcal{A}}$, yielding (3.3). □

Remark 3.2 (Shadow connection interpretation). The KZB connection is the genus-1 instance of the modular shadow connection $\nabla^{\mathrm{hol}}_{g,n}$ on $\overline{\mathcal{M}}_{g,n}$: at genus 0 it gives the KZ connection, at genus 1 the KZB connection, and at genus ≥ 2 it produces new flat connections on configuration spaces of higher-genus curves.

4 Face 5: the elliptic r-matrix

At genus 0, Face 5 identifies the collision residue with the Drinfeld r-matrix $\Omega/((k+h^\vee)z)$. At genus 1, the rational r-matrix is replaced by the classical elliptic r-matrix of Belavin [1] and Belavin–Drinfeld [2].

Theorem 4.1 (Face 5: elliptic r-matrix)**.** *For $\mathcal{A} = \widehat{\mathfrak{g}}_k$ with $\mathfrak{g}$ simple and $k \neq -h^\vee$, the genus-1 collision residue equals the classical elliptic r-matrix up to the standard level normalization:*

$$r^{(1)}_{\widehat{\mathfrak{g}}_k}(z,\tau) \;=\; \frac{1}{k+h^\vee}\,r^{\mathrm{ell}}_{\mathfrak{g}}(z,\tau). \tag{4.1}$$

For $\mathfrak{g} = \mathfrak{sl}_2$, the Belavin r-matrix is

$$r^{\mathrm{ell}}_{\mathfrak{sl}_2}(z,\tau) \;=\; \zeta_\tau(z)\cdot\frac{H\otimes H}{2} \;+\; \phi_+(z,\tau)\,E\otimes F \;+\; \phi_-(z,\tau)\,F\otimes E, \tag{4.2}$$

where $\{H,E,F\}$ is the standard basis of $\mathfrak{sl}_2$ and the theta-function ratios are

$$\phi_\pm(z,\tau) \;=\; \frac{\theta_1'(0\mid\tau)\,\theta_1(z\pm\frac{1}{2}\mid\tau)}{\theta_1(z\mid\tau)\,\theta_1(\pm\frac{1}{2}\mid\tau)}. \tag{4.3}$$

The elliptic r-matrix satisfies the classical Yang–Baxter equation on E_τ:

$$[r^{\mathrm{ell}}_{12}(z_1-z_2),\,r^{\mathrm{ell}}_{13}(z_1-z_3)] + [r^{\mathrm{ell}}_{12}(z_1-z_2),\,r^{\mathrm{ell}}_{23}(z_2-z_3)] + [r^{\mathrm{ell}}_{13}(z_1-z_3),\,r^{\mathrm{ell}}_{23}(z_2-z_3)] = 0. \tag{4.4}$$

Proof. For the Cartan channel: the Kac–Moody OPE with $c_0 = \Omega/(k+h^\vee)$ gives, after genus-1 regularization, $\zeta_\tau(z)\cdot H\otimes H/2$ in the Cartan direction.

For the root channels: on E_τ, the chiral fields J^E, J^F associated to the positive and negative roots satisfy twisted boundary conditions under B-cycle monodromy. The collision residue in the root channels acquires the theta-function ratios $\phi_\pm(z,\tau)$ (4.3), which are the unique meromorphic functions on $\mathbb{C}$ with a simple pole at $z=0$ matching $1/z$ and with B-cycle quasi-periodicity dictated by the root lattice [1].

The classical Yang–Baxter equation (4.4) follows from the collision-depth-2 projection of the Maurer–Cartan equation at genus 1: the infinitesimal braid relation $[\Omega_{ij}, \Omega_{ik}+\Omega_{jk}]=0$ is replaced by its elliptic counterpart. The overall identification (4.1) follows from Theorem 2.4 combined with uniqueness of the classical elliptic r-matrix [2]. □

Remark 4.2 (Higher rank). For $\mathfrak{g}=\mathfrak{sl}_N$ with $N\geq 3$, the Belavin elliptic r-matrix is a matrix-valued meromorphic function on E_τ with poles at $z=0$ whose residue is the $\mathfrak{sl}_N$ Casimir. The collision residue produces this r-matrix by the same mechanism: elliptic regularization of the Cartan and root channels, with theta-function ratios $\phi_\alpha(z,\tau)$ indexed by the roots α. The identification (4.1) holds for all simple $\mathfrak{g}$.

5 Face 7: the elliptic Gaudin model

At genus 0, Face 7 identifies the collision residue with the Gaudin Hamiltonians $H_i^{\mathrm{Gaudin}} = \sum_{j\neq i}\Omega_{ij}/(z_i-z_j)$. At genus 1, the Gaudin model acquires elliptic structure.

Theorem 5.1 (Face 7: elliptic Gaudin Hamiltonians)**.** *For $\mathcal{A}=\widehat{\mathfrak{g}}_k$, the elliptic Gaudin Hamiltonians on* $\mathrm{Conf}_n(E_\tau)$ *are*

$$H_i^{\mathrm{ell}} \;=\; \frac{1}{k+h^\vee}\sum_{j\neq i}\Omega_{ij}\,\zeta_\tau(z_i-z_j) \;=\; \sum_{j\neq i}\big[r^{(1)}_{\widehat{\mathfrak{g}}_k}(z_i-z_j,\tau)\big]_{ij}. \tag{5.1}$$

They mutually commute: $[H_i^{\mathrm{ell}}, H_j^{\mathrm{ell}}]=0$ *for all* i,j.

Proof. The identification of H_i^{ell} with the collision residue is immediate from Theorem 2.4: $r^{(1)}_{\widehat{\mathfrak{g}}_k}(z,\tau)=\Omega\,\zeta_\tau(z)/(k+h^\vee)$.

For commutativity, the Maurer–Cartan equation at genus 1 gives $D_{\mathcal{A}}\Theta_{\mathcal{A}}+\frac{1}{2}[\Theta_{\mathcal{A}},\Theta_{\mathcal{A}}]=0$ restricted to the collision-depth-2 stratum of $\overline{\mathcal{M}}_{1,n}$. Expanding the bracket and projecting onto the (i,j) and (i,k) channels produces the elliptic infinitesimal braid relation

$$[\zeta_\tau(z_{ij})\,\Omega_{ij},\;\; \zeta_\tau(z_{ik})\,\Omega_{ik}+\zeta_\tau(z_{jk})\,\Omega_{jk}] \;=\; 0, \tag{5.2}$$

which follows from the classical elliptic YBE (4.4) by specializing to three points and evaluating on the Casimir. The commutativity $[H_i^{\mathrm{ell}}, H_j^{\mathrm{ell}}]=0$ is the n-point consequence of (5.2). □

Remark 5.2 (Hitchin system). The elliptic Gaudin model is the genus-1 instance of the Hitchin integrable system on $\mathrm{Bun}_G(E_\tau)$. The Hitchin Hamiltonians are sections of a line bundle on the base of the Hitchin fibration; at genus 1, the base is the affine space of elliptic spectral curves, and the Hitchin Hamiltonians restrict to the elliptic Gaudin Hamiltonians on the trivial G-bundle. At genus $g\geq 2$, the collision residue on Σ_g produces the Hitchin Hamiltonians on $\mathrm{Bun}_G(\Sigma_g)$; this is the structural target of the holographic programme [8].

6 Class M at genus 1: higher-order elliptic operators

For affine Kac–Moody algebras (class L), the collision expansion terminates at c_0: the collision depth is $k_{\max} = 1$, and only the ζ_τ-term contributes. For class M algebras (Virasoro, $\mathcal{W}_N$), the OPE has poles of higher order, and the collision expansion involves $\wp$, $\wp'$, and higher derivatives of the Weierstrass functions.

6.1 Virasoro at genus 1

The Virasoro algebra has maximal OPE pole order $p_{\max} = 4$ (from the $T(z)\,T(w)$ OPE). The bar propagator absorbs one order, giving collision depth $k_{\max} = 3$. The genus-0 collision residue is

$$r_{\mathrm{Vir}_c}(z) \;=\; \frac{c/2}{z^3} \;+\; \frac{2T}{z}, \tag{6.1}$$

with $c_0 = 2T$, $c_1 = 0$ (the z^{-2} pole is absent from the r-matrix by a parity argument: the $d\log$ extraction sends z^{-2n} to $z^{-(2n-1)}$), and $c_2 = c/2$.

Theorem 6.1 (Virasoro genus-1 collision residue)**.** *The genus-1 collision residue of* Vir_c *is*

$$r^{(1)}_{\mathrm{Vir}_c}(z,\tau) \;=\; 2T\cdot\zeta_\tau(z) \;-\; \frac{c}{4}\,\wp'(z,\tau). \tag{6.2}$$

This is an elliptic differential operator of order 2 *on conformal blocks on* E_τ*: the* ζ_τ*-term acts by multiplication, while the* $\wp'$*-term acts as a second-order differential operator through the stress-energy Ward identity.*

Proof. The genus-0 collision expansion has $c_0 = 2T$, $c_1 = 0$, $c_2 = c/2$. By Theorem 2.4:

$$r^{(1)}_{\mathrm{Vir}_c}(z,\tau) \;=\; c_0\,\zeta_\tau(z) + c_1\,\wp(z,\tau) - \tfrac{1}{2}\,c_2\,\wp'(z,\tau).$$

Since $c_1 = 0$, the $\wp$-term vanishes, and the result is (6.2). The Laurent expansion near $z = 0$ confirms consistency: $2T/z - (c/4)\cdot(-2/z^3 + O(z)) = (c/2)/z^3 + 2T/z + O(1)$, matching $r_{\mathrm{Vir}_c}(z)$ at leading order.

The $\wp'$-term acts as a second-order differential operator because the Virasoro OPE at order $(z-w)^{-4}$ involves $(c/2)\cdot\mathrm{id}$, which through the Ward identity produces a second derivative ∂_z^2 on the correlator; the elliptic regularization replaces $1/z^4$ by $-\wp'(z,\tau)/2$. □

Remark 6.2 (The $c_1 = 0$ vanishing). The vanishing of c_1 is a parity phenomenon: the OPE mode $T_{(2)}T = \partial T$ is a total derivative, and its contribution to the r-matrix vanishes after $d\log$-extraction. In the elliptic regularization, this means the $\wp(z,\tau)$-term is absent. The first non-trivial correction beyond ζ_τ involves $\wp'$, not $\wp$.

6.2 $\mathcal{W}_N$ at genus 1

The $\mathcal{W}_N$ algebra has generators $W_2 = T, W_3, \ldots, W_N$ of conformal weights $2, 3, \ldots, N$. The maximal OPE pole order among generating fields is $p_{\max} = 2N$ (from the W_N-W_N OPE), giving collision depth $k_{\max} = 2N - 1$.

Theorem 6.3 ($\mathcal{W}_N$ genus-1 collision residue)**.** *The genus-1 collision residue of $\mathcal{W}_N$ involves the Weierstrass functions up to $\wp^{(2N-3)}$:*

$$r^{(1)}_{\mathcal{W}_N}(z,\tau) \;=\; c_0\,\zeta_\tau(z) \;+\; \sum_{\substack{n=1\\ c_n\neq 0}}^{2N-2} \frac{(-1)^{n-1}}{n!}\, c_n \cdot \wp^{(n-1)}(z,\tau), \tag{6.3}$$

where c_n are the OPE-mode collision coefficients. The highest-order term is an elliptic differential operator of order $2N-2$. For $\mathcal{W}_3$: the collision expansion involves $\zeta_\tau, \wp, \wp', \wp'', \wp'''$, producing fourth-order elliptic differential operators.

Proof. Direct specialization of Theorem 2.4 to the $\mathcal{W}_N$ OPE data with $p_{\max} = 2N$: the collision depth $k_{\max} = 2N-1$ means the genus-0 expansion $r_{\mathcal{W}_N}(z) = \sum_{n=0}^{2N-2} c_n/z^{n+1}$ has $2N-1$ terms, each regularized by $1/z \mapsto \zeta_\tau(z)$ and $1/z^{n+1} \mapsto (-1)^{n-1}\wp^{(n-1)}(z)/n!$ for $n \geq 1$. □

Remark 6.4 (Genuinely new operators). The elliptic differential operators produced by the class M collision residue are not present in the existing integrable systems literature. The Hitchin system on $\mathrm{Bun}_G(E_\tau)$ involves only the ζ_τ-term (collision depth 1, corresponding to the Casimir of $\mathfrak{g}$). The KZB connection involves ζ_τ and $\wp$ (depths 1 and 2). The class M collision residue at depth ≥ 3 produces operators involving $\wp', \wp'', \ldots, \wp^{(2N-3)}$ that arise from higher OPE poles specific to $\mathcal{W}$-algebras and have no counterpart in the theory of affine Kac–Moody algebras.

7 Degeneration to genus 0

Theorem 7.1 (Degeneration)**.** *As $\mathrm{Im}(\tau) \to \infty$ (equivalently, $q = e^{2\pi i\tau} \to 0$), all seven genus-1 faces degenerate to their genus-0 counterparts:*

(i) (Propagator.) $\zeta_\tau(z) \to \pi\cot(\pi z) \to 1/z + O(z)$.

(ii) (Weierstrass functions.) $\wp(z,\tau) \to \pi^2/\sin^2(\pi z) \to 1/z^2 + O(1)$, *and* $\wp^{(m)}(z,\tau) \to (-1)^m(m+1)!/z^{m+2} + O(z^{-m})$.

(iii) (KZB $\to$ KZ.) *The dz-component degenerates to the KZ connection: $\zeta_\tau(z_{ij}) \to 1/z_{ij}$. The $d\tau$-component has no genus-0 counterpart.*

(iv) (Elliptic $\to$ rational r-matrix.) $r^{\mathrm{ell}}_{\mathfrak{g}}(z,\tau)/(k+h^\vee) \to k\,\Omega/((k+h^\vee)z) = \Omega/((k+h^\vee)z)$ *through the intermediate trigonometric r-matrix: $r^{\mathrm{ell}} \to r^{\mathrm{trig}} \to r^{\mathrm{rat}}$.*

(v) (Elliptic $\to$ rational Gaudin.) $H_i^{\mathrm{ell}} \to H_i^{\mathrm{Gaudin}} = \sum_{j\neq i}\Omega_{ij}/(z_i - z_j)$.

Proof. As $q \to 0$, the Weierstrass functions have the standard degenerations. The zeta function: $\zeta_\tau(z) = \pi\cot(\pi z) + O(q)$, giving $1/z + O(z)$ for $|z| < 1$. The $\wp$-function: $\wp(z,\tau) = \pi^2/\sin^2(\pi z) - 2\pi^2 E_2(\tau)/3 + O(q)$, giving $1/z^2 + O(1)$. Higher derivatives degenerate analogously.

The KZB degeneration: with $\zeta_\tau(z) \to 1/z$, the dz-component (3.2) becomes $\sum_{j\neq i}\Omega_{ij}/((k+h^\vee)z_{ij})\,dz_{ij}$, which is the KZ connection.

The elliptic r-matrix degeneration: as $q \to 0$, $\phi_\pm(z,\tau) \to \pi/\sin(\pi z) + O(q)$, recovering the trigonometric r-matrix. A further limit $z \mapsto z/L$, $L \to \infty$ gives $\pi/\sin(\pi z/L) \to L/z$, recovering the rational r-matrix $\Omega/((k+h^\vee)z)$ in the KZ normalization. Both limits are continuous in the topology of formal Laurent series. □

The degeneration theorem provides the consistency check between genus-0 and genus-1: every genus-1 face is a continuous deformation of the corresponding genus-0 face, with deformation parameter q. At $q = 0$, the elliptic curve degenerates to the nodal rational curve, and all seven identifications collapse to their genus-0 ancestors.

8 Computational verification

The genus-1 seven-face identifications are verified by the compute engine `theorem_genus1_seven_faces_engine.py` which implements the following tests.

(1) **Elliptic regularization.** For each standard family $\mathcal{A}$ in the landscape census, verify that the genus-1 collision residue (2.5) matches the elliptic regularization of the genus-0 collision expansion at $O(q^0)$ to machine precision.

(2) **KZB identification.** For $\mathcal{A} = \widehat{\mathfrak{sl}}_2$ at levels $k = 1, 2, 3, 5, 10$, verify that the dz-component of the KZB connection equals the collision residue, and that the $d\tau$-component equals $-\partial_z r^{(1)}_{\mathcal{A}}$.

(3) **Elliptic CYBE.** Verify the classical elliptic Yang–Baxter equation (4.4) numerically for $\mathfrak{sl}_2$ at 20 random evaluation points $(z_1, z_2, z_3) \in E_\tau^3$ and 5 values of τ.

(4) **Gaudin commutativity.** Verify $[H_i^{\mathrm{ell}}, H_j^{\mathrm{ell}}] = 0$ for $\mathfrak{sl}_2$ on $\mathrm{Conf}_3(E_\tau)$ at 10 random configurations and 5 values of τ.

(5) **Degeneration.** Verify that $\|r^{(1)}_{\mathcal{A}}(z,\tau) - r_{\mathcal{A}}(z)\|/\|r_{\mathcal{A}}(z)\| < 10^{-6}$ for $\mathrm{Im}(\tau) \geq 12$ across all standard families.

(6) **Class M structure.** For Vir_c at $c = 1, 1/2, 25, 26$, verify (6.2) and confirm the $c_1 = 0$ vanishing. For $\mathcal{W}_3$ at $c = 2, 50, 98$, verify (6.3) and confirm terms through $\wp'''$.

9 The Verlinde formula from ordered chiral homology

At integer level $k \in \mathbb{Z}_{\geq 1}$, the ordered chiral homology of $\widehat{\mathfrak{g}}_k$ on E_τ recovers the Verlinde partition function. The connection to the seven-face programme is direct: the KZB connection of Theorem 3.1 has finite monodromy at integer level, and the trace of the B-cycle monodromy over the space of conformal blocks recovers the Verlinde formula.

Proposition 9.1 (Verlinde from genus-1 collision residue)**.** *For $\mathcal{A} = \widehat{\mathfrak{g}}_k$ at integer level $k \geq 1$, the B-cycle monodromy of the KZB connection is*

$$\mathcal{M}_B \;=\; \exp\Big(\frac{2\eta_\tau}{k+h^\vee}\,\Omega\Big) \;=\; \exp\big(2\eta_\tau \cdot c_0\big), \tag{9.1}$$

where $c_0 = \Omega/(k+h^\vee)$ is the degree-0 collision coefficient. The quantum group parameter is $q = \exp(2\pi i/(k+h^\vee))$, and the Verlinde partition function at genus 1 is

$$Z_1(\widehat{\mathfrak{g}}_k) \;=\; \sum_{\lambda \in P_+^k} |S_{0\lambda}|^0 \;=\; |P_+^k|, \tag{9.2}$$

where P_+^k is the set of integrable highest weights at level k and $S_{0\lambda}$ are the modular S-matrix entries. At higher genus g, the Verlinde formula $Z_g = \sum_\lambda S_{0\lambda}^{2-2g}$ is recovered from the ordered chiral homology via handle attachment and separating factorization of the ordered bar complex.

Remark 9.2 (The B-cycle monodromy and the quantum group). The quasi-periodicity (2.7) of the genus-1 collision residue is the geometric origin of the quantum group parameter. For affine $\widehat{\mathfrak{sl}}_{2,k}$: the B-cycle monodromy eigenvalues are $q^{j(j+1)}$ for $j = 0, 1/2, \ldots, k/2$, where $q = e^{2\pi i/(k+2)}$ is a root of unity. The passage from the E_1 collision residue $r^{(1)}(z,\tau)$ to the quantum group $U_q(\mathfrak{g})$ is the B-cycle monodromy of the ordered bar complex: the spectral parameter z (genus-0 data) becomes the quantum group parameter q (genus-1 data) through exponentiation of the quasi-period.

10 The critical level

The seven-face programme at genus 1 has a distinguished boundary: the critical level $k = -h^\vee$, where the Sugawara construction degenerates. The collision residue and the KZB connection undergo a qualitative change at this value.

Theorem 10.1 (Critical level jump at genus 1)**.** *For $\mathcal{A} = \widehat{\mathfrak{g}}_k$ at the critical level $k = -h^\vee$:*

(i) *The shadow invariant vanishes: $\kappa(\widehat{\mathfrak{g}}_{-h^\vee}) = 0$. All monodromy eigenvalues of the genus-1 collision residue become integers.*

(ii) *The degree-2 ordered chiral homology on $V \otimes V$ doubles: $\dim H^1_{\mathrm{ord},2}$ jumps from 4 (generic level) to 8 (critical level) for $\mathfrak{g} = \mathfrak{sl}_2$ (finite computation, $\chi_{\mathrm{ord},2} = -4$ conserved). This is distinct from the full symmetric bar cohomology at critical level, addressed in (iii).*

(iii) *The FULL symmetric bar cohomology at critical level computes the algebra of differential forms on the space of $\mathfrak{g}^\vee$-opers: $H^n(B^\Sigma(\widehat{\mathfrak{g}}_{-h^\vee})) \cong \Omega^n(\mathrm{Op}_{\mathfrak{g}^\vee}(D))$ in every degree $n \geq 0$ (infinite-dimensional; Feigin–Frenkel 1992, Frenkel–Teleman 2006).*

(iv) *The genus-1 collision residue degenerates: the KZ normalization $r^{(1)}(z,\tau) = \Omega\,\zeta_\tau(z)/(k + h^\vee)$ has a pole at $k = -h^\vee$, and the KZB connection becomes singular. The critical level requires a separate regularization via the Feigin–Frenkel center [5].*

Remark 10.2 (Koszulness at critical level). At generic level $k \neq -h^\vee$, the affine Kac–Moody algebra $\widehat{\mathfrak{g}}_k$ is chirally Koszul, and the seven-face programme applies without modification. At the critical level $k = -h^\vee$, chirally Koszulness fails: the center of $\widehat{\mathfrak{g}}_{-h^\vee}$ jumps (it becomes the Feigin–Frenkel center, isomorphic to the algebra of functions on the space of opers), and the bar complex acquires additional cohomology. The seven faces degenerate: the Gaudin Hamiltonians become the Feigin–Frenkel higher Gaudin Hamiltonians, and the r-matrix limit requires the Langlands dual group ${}^LG$ via the geometric Langlands correspondence. The critical level is not an obstruction to the programme; it is a boundary where new structure emerges.

Remark 10.3 (Spectral sequence at critical level). The passage from generic to critical level is controlled by a spectral sequence with $d_1 = \lambda \cdot [\delta]$, where $\lambda = k + h^\vee$ is the level shift and δ is the boundary class. At $\lambda = 0$ (critical level), d_1 vanishes, and the spectral sequence degenerates at E_1: the bar cohomology jumps because the differential that would kill the extra classes is absent. This is the mechanism by which the degree-2 ordered chiral homology $H^1_{\mathrm{ord},2}$ doubles from 4 to 8 for $\mathfrak{sl}_2$. For the FULL symmetric bar cohomology at critical level, the d_1 vanishing lets the entire de Rham algebra $\Omega^*(\mathrm{Op}_{\mathfrak{sl}_2}(D))$ survive as bar cohomology, which is infinite-dimensional.

References

[1] A. A. Belavin, *Dynamical symmetry of integrable quantum systems*, Nucl. Phys. B **180** (1981), 189–200.

[2] A. A. Belavin and V. G. Drinfeld, *Solutions of the classical Yang–Baxter equation for simple Lie algebras*, Funct. Anal. Appl. **16** (1982), 159–180.

[3] D. Bernard, *On the Wess–Zumino–Witten models on the torus*, Nucl. Phys. B **303** (1988), 77–93.

[4] T. Dimofte, W. Niu, and A. Py, *Meromorphic tensor products and line operators in* $3d$ *TQFT*, preprint, 2025.

[5] B. Feigin, E. Frenkel, and N. Reshetikhin, *Gaudin model, Bethe ansatz and critical level*, Comm. Math. Phys. **166** (1994), 27–62.

[6] O. Khan and M. Zeng, *Poisson vertex algebras and* $3d$ *sigma models*, preprint, 2025.

[7] D. Bernard, *On the Wess–Zumino–Witten models on Riemann surfaces*, Comm. Math. Phys. **124** (1989), 315–340; V. G. Knizhnik and A. B. Zamolodchikov, *Current algebra and Wess–Zumino model in two dimensions*, Nucl. Phys. B **247** (1984), 83–103.

[8] R. Lorgat, *Modular Koszul Duality*, preprint, 2026.

[9] M. A. Semenov-Tian-Shansky, *What is a classical* r*-matrix?*, Funct. Anal. Appl. **17** (1983), 259–272.